

BILL OF MATERIALS

LoRa-Based Pet Tracking Collar Unit

Currency: Indian Rupee (₹) | Quantity basis: 1 unit (prototype)

#	Ref. Des.	Component	Description / Value	Package	Qty	Unit Cost (₹)	Total Cost (₹)
1. USB Type-C & CH340E Interface Section							
1	P1	USB-C Receptacle	USB_C_Plug_USB2.0, 6-pin SMD	USB-C	1	28	28
2	U6	CH340E	USB to UART Bridge IC	MSOP-10	1	49	49
3	R7, R8	Resistor	22 Ω, USB D+/D- series	0603	2	1	2
4	C3	Ceramic Capacitor	100 nF, decoupling	0603	1	1	1
5	C4	Ceramic Capacitor	10 μF, decoupling	0805	1	4	4
2. Battery & Voltage Controller Section							
6	U5	TP4056-42-ESOP8	Li-ion/LiPo Linear Charger IC	ESOP-8	1	28	28
7	U4	AMS1117-3.3	3.3 V LDO Voltage Regulator	SOT-223	1	17	17
8	D1, D2	Diode	Schottky / status indicator diode	SOD-123	2	6	12
9	R1–R6	Resistor	Charge-current set / status, misc.	0603	6	1	6
10	C1, C2	Capacitor	10 μF & 100 nF, input/output filter	0805/0603	2	3	6
11	BT1	Li-ion/LiPo Battery Cell	3.7 V single-cell rechargeable	Cell	1	350	350
3. Reset System Section							
12	R9	Resistor	10 kΩ pull-up	0603	1	1	1
13	SW1	SW_Push_Dual	Tactile push-button switch	SMD	1	11	11
14	C8	Ceramic Capacitor	100 nF, debounce	0603	1	1	1
4. GPS Section (MAX-M10S)							
15	U2	MAX-M10S	u-blox GNSS/GPS Module	LGA-18	1	1260	1260
16	AE2	GNSS Antenna	Passive chip/patch GPS antenna	SMD	1	210	210
5. MCU / LoRa Section (RAK3172)							
17	U1	RAK3172-xx-B-SM-xl	STM32WLE5-based LoRa Module	SMD-32	1	980	980
18	C5, C6, C7	Capacitor	100 nF x2, 10 μF x1, decoupling	0603/0805	3	3	9
19	E1	ANT-868-CHIP-T	868 MHz Monopole Ceramic Chip Antenna	SMD	1	168	168
6. Accelerometer Sensor Section							
20	U3	LIS2DE12	3-Axis Digital Accelerometer	LGA-12	1	133	133
7. PCB & Mechanical							
21	—	Custom PCB	2-layer FR4, fabrication (per unit, low volume)	—	1	420	420
ESTIMATED GRAND TOTAL (per unit)							₹ 3,696

Notes

- Unit costs are approximate, indicative retail/distributor prices in Indian Rupees (₹) for single-unit / prototype quantities, sourced from Indian and international component distributors as of September 2026. Actual prices vary with vendor, package variant, and exchange rate.
- Prices exclude GST, shipping/freight, PCB assembly (SMT) labour, and enclosure/mechanical hardware not shown on the schematic.
- Costs of key modules (RAK3172 LoRa module, MAX-M10S GPS module) and the battery cell drop significantly at volumes of 1000+ units — bulk/OEM pricing should be requested from distributors for production runs.